Lecture 8
The Network Layer
Control Plane and ICMP

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Subjects of today:

- The Core of the Control Plane
- Intra Autonomous System Routing
- Inter Autonomous System Routing
- Software Defined Networking & Generalized Forwarding in Brief
- Internet Control Message Protocol

8.1 The Core of the Control Plane

Network-layer functions

Recall: two network-layer functions:

- **Forwarding**: move packets from router's input to appropriate router output → Data plane
- **Routing**: determine route taken by packets from source to destination → Control plane

Two approaches to structuring network control plane:

1. per-router control (traditional)
2. logically centralized control (software defined networking)

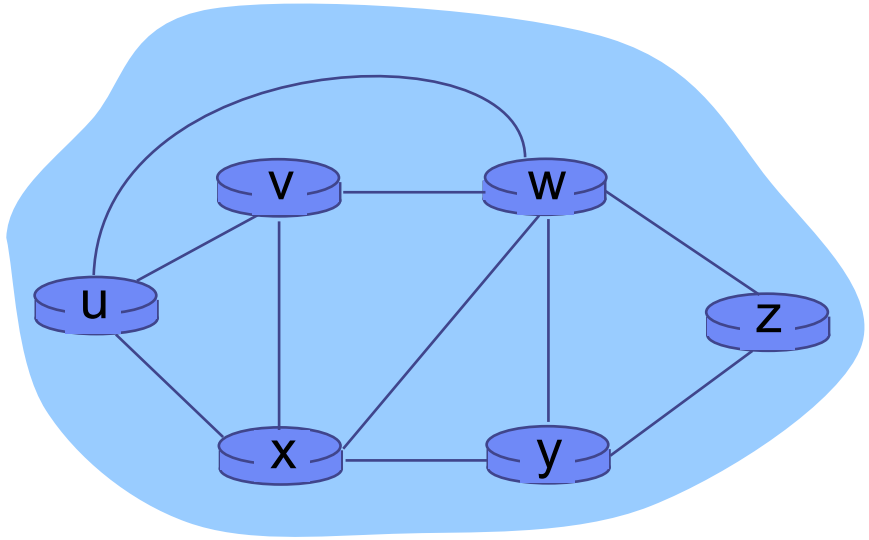
Routing protocols

Goal: Determine *good paths* from sending hosts to receiving host, through network of routers

- *Path*: Sequence of routers - that packets will traverse in going from given initial source host to given final destination host
- "*Good*": Least *cost*

- Cost could always be 1, or inversely related to bandwidth, or related to congestion. But, also fees, delay, quality and politics.

Graph Abstraction of the Network



Graph: G = (N,E)

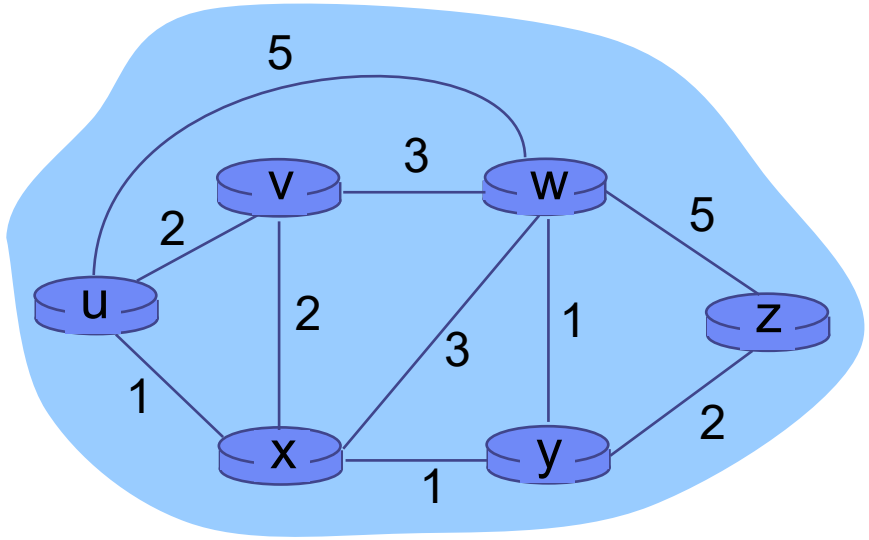
N = set of routers = { u, v, w, x, y, z }

E = set of links ={ (u,v), (u,x), (v,x), (v,w), (x,w), (x,y), (w,y), (w,z), (y,z) }

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Graph Abstraction: Costs

c(n,n') = cost of link (n,n')
e.g., c(w,z) = 5



cost of path $(n_1, n_2, n_3, \ldots, n_p) = c(n_1,n_2) + c(n_2,n_3) + \ldots + c(n_{p-1},n_p)$

Key question: what is the least-cost path between u and z ?
Routing algorithm: algorithm that finds that least cost path

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Routing Algorithm Classification

Q: global or decentralized information?

Global:

- all routers have complete topology, link cost info
- "link state" algorithms

Decentralized:

- router knows physically-connected neighbors, link costs to neighbors
- iterative process of computation, exchange of info with neighbors
- "distance vector" algorithms

Q: static or dynamic?

Static:

- routes change slowly over time

Dynamic:

- routes change more quickly
  - periodic update
  - in response to link cost changes

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A Link-state Routing Algorithm

Dijkstra's algorithm

- net topology, link costs known to all nodes
  - accomplished via "link state broadcast"
  - all nodes have same info
- computes least cost paths from one node ("source") to all other nodes
  - gives *forwarding table* for that node
- iterative: after k iterations, know least cost path to k dest.'s

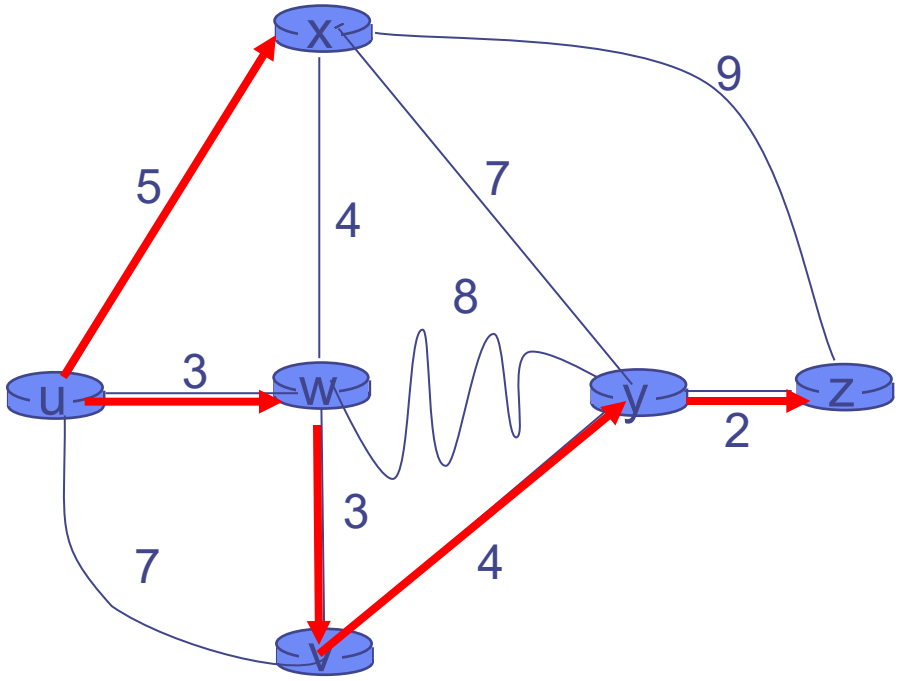
Notation:

- c(x,y): link cost from node x to y; = ∞ if not direct neighbors
- D(v): current value of cost of path from source to dest. v
- p(v): predecessor node along path from source to v
- N': set of nodes whose least cost path definitively known

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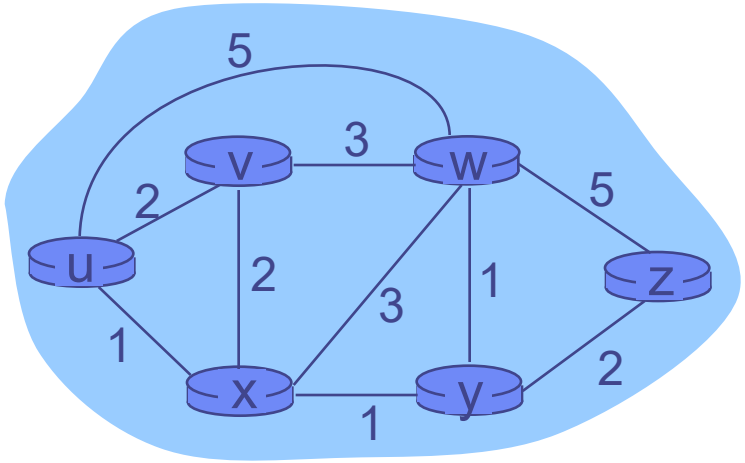
Dijkstra's algorithm: Example

Step	N'	D(**v**) p(v)	D(**w**) p(w)	D(**x**) p(x)	D(**y**) p(y)	D(**z**) p(z)
0	u	7,u	3,u	5,u	∞	∞
1	uw	6,w		5,u	11,w	∞
2	uwx	6,w			11,w	14,x
3	uwxv				10,v	14,x
4	uwxvy					12,y
5	uwxvyz					



Dijkstra's algorithm: another example

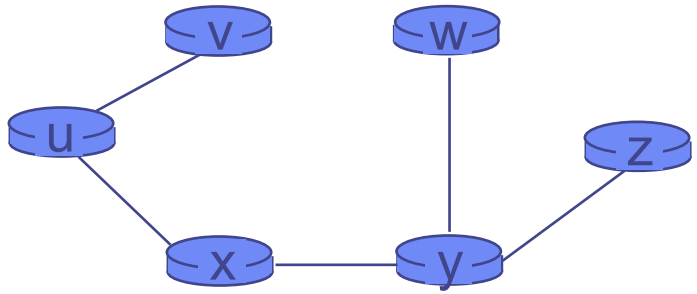
Step	N'	D(v),p(v)	D(w),p(w)	D(x),p(x)	D(y),p(y)	D(z),p(z)
0	u	2,u	5,u	1,u	∞	∞
1	ux	2,u	4,x		2,x	∞
2	uxy	2,u	3,y			4,y
3	uxyv		3,y			4,y
4	uxyvw					4,y
5	uxyvwz					



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Dijkstra's algorithm: example (2)

Resulting shortest-path tree from u:



Resulting forwarding table in **node u**:

destination	link
v	(u,v)
x	(u,x)
y	(u,x)
w	(u,x)
z	(u,x)

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A Distance Vector Routing Algorithm

Bellman-Ford equation (dynamic programming)

let

$d_x(y)$:= cost of least-cost path from x to y

then

$d_x(y) = min_v \{c(x,v) + d_v(y)\}$

cost from neighbor v to destination y

cost to neighbor v

min taken over all neighbors v of x

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Bellman-Ford Example

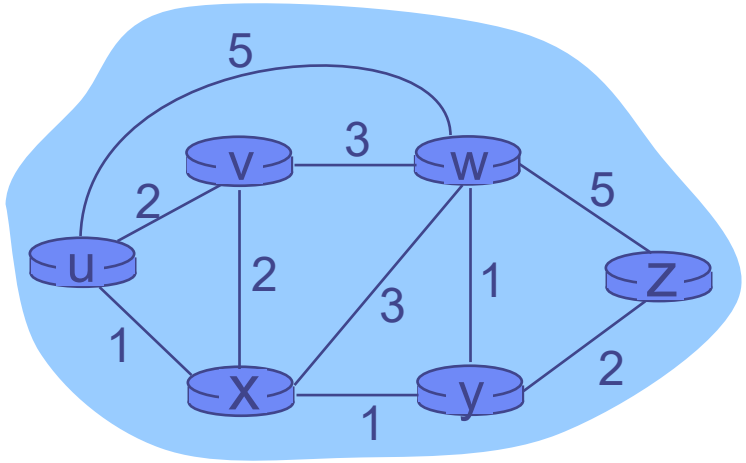
$d_v(z) = 5$, $d_x(z) = 3$, $d_w(z) = 3$

B-F equation says:

$$d_u(z) = \min \{ c(u,v) + d_v(z), \; c(u,x) + d_x(z), \; c(u,w) + d_w(z) \}$$

$$= \min \{ 2 + 5, 1 + 3, 5 + 3 \}$$

$$= 4$$



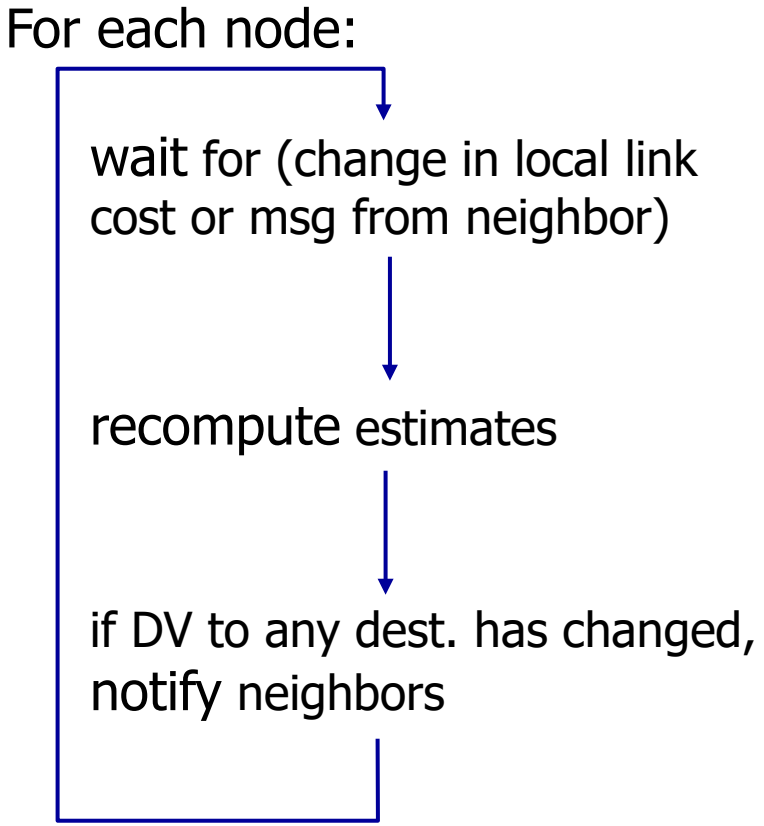
Distance Vector Algorithm Strategy

Iterative, asynchronous: each local iteration caused by:

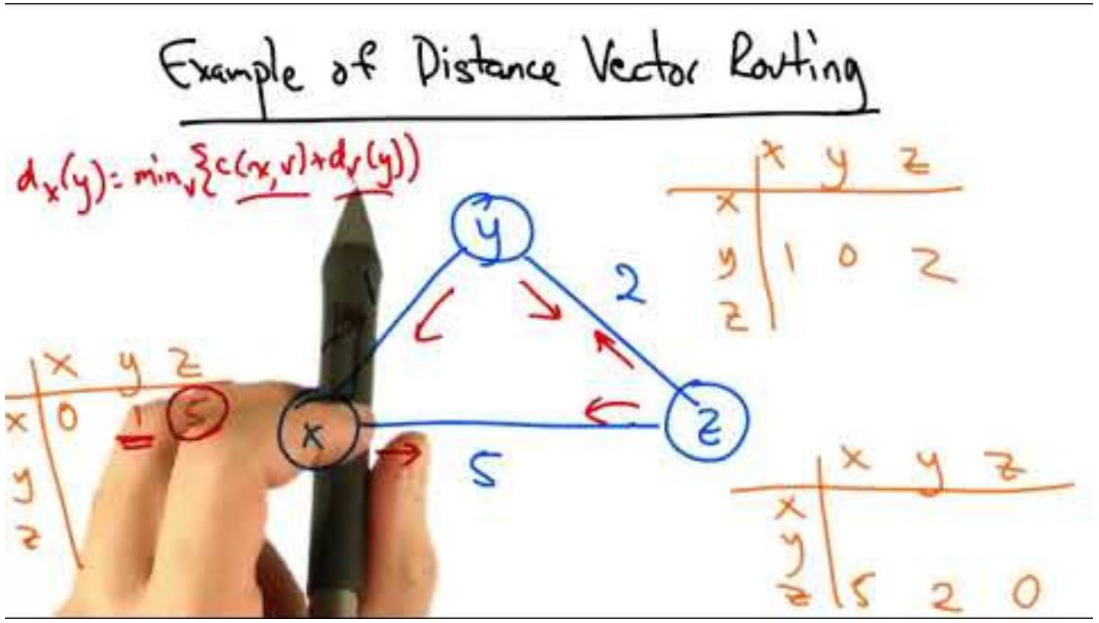
- local link cost change
- Distance vector (DV) update message from neighbor

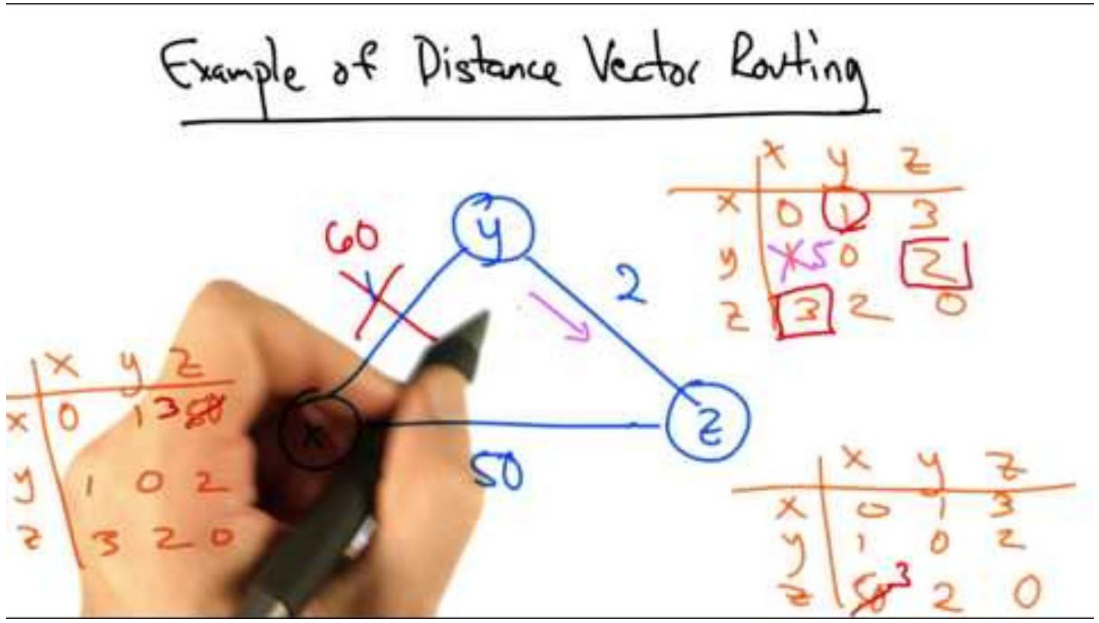
Distributed:

- each node notifies neighbors only when its DV changes
  - neighbors then notify their neighbors ONLY if necessary



Distance Vector Algorithm Example 2





Comparison of LS and DV algorithms

Message complexity

- **LS:** with n nodes, E links, O(nE) msgs sent
- **DV:** exchange between neighbors only
  - convergence time varies

Speed of convergence

- **LS:** $O(n^2)$ algorithm
  - may have oscillations
- **DV:** convergence time varies
  - may be routing loops
  - count-to-infinity problem

Robustness: what happens if router malfunctions?

LS:

- node can advertise incorrect *link* cost
- each node computes only its *own* table

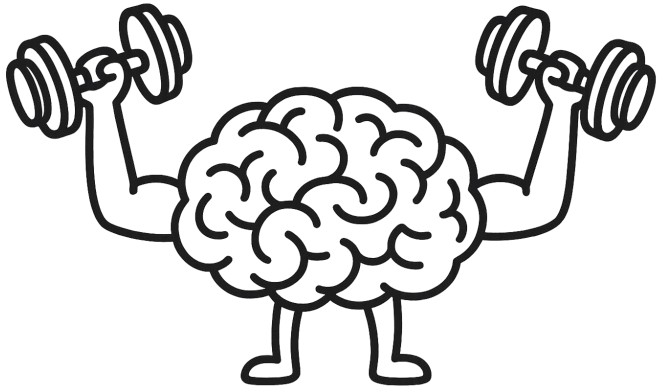
DV:

- DV node can advertise incorrect *path* cost (black-holing)
- each node's table used by others
  - error propagate through network

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Routing algorithm: Exercises

1. Go to: http://gaia.cs.umass.edu/kurose_ross/interactive/
2. Complete the exercise called - Dijkstra's Link State Algorithm
3. Complete the exercise called - Bellman Ford Distance Vector Algorithm



8.2 Intra Autonomous System Routing

Making Routing Scalable

Our routing study thus far - idealized

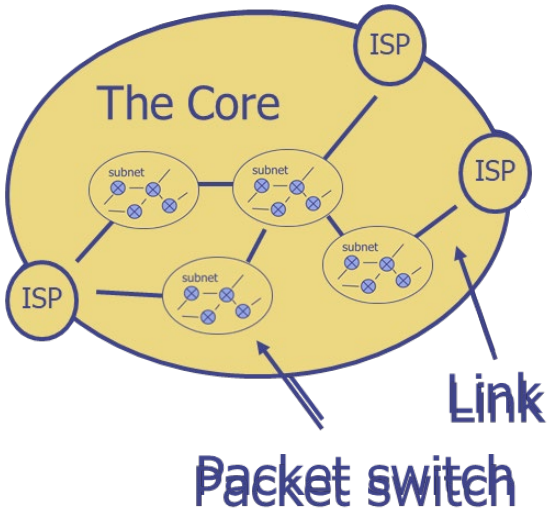
- all routers identical
- network “flat”
  … *not* true in practice

Scale consideration: with billions of destinations:

- can’t store all destinations in routing tables!
- routing table exchange would swamp links!

Administrative autonomy consideration

- internet = network of networks
- each network admin may want to control routing in its own network



Internet approach to scalable routing

Aggregate routers into regions known as "autonomous systems" (AS) (a.k.a. "domains")

Intra-AS routing

- Routing among hosts, routers in same AS ("network")
- All routers in AS must run *same* intra-domain protocol
- Routers in *different* AS can run *different* intra-domain routing protocol
- **Gateway router:** at "edge" of its own AS, has link(s) to router(s) in other AS'es

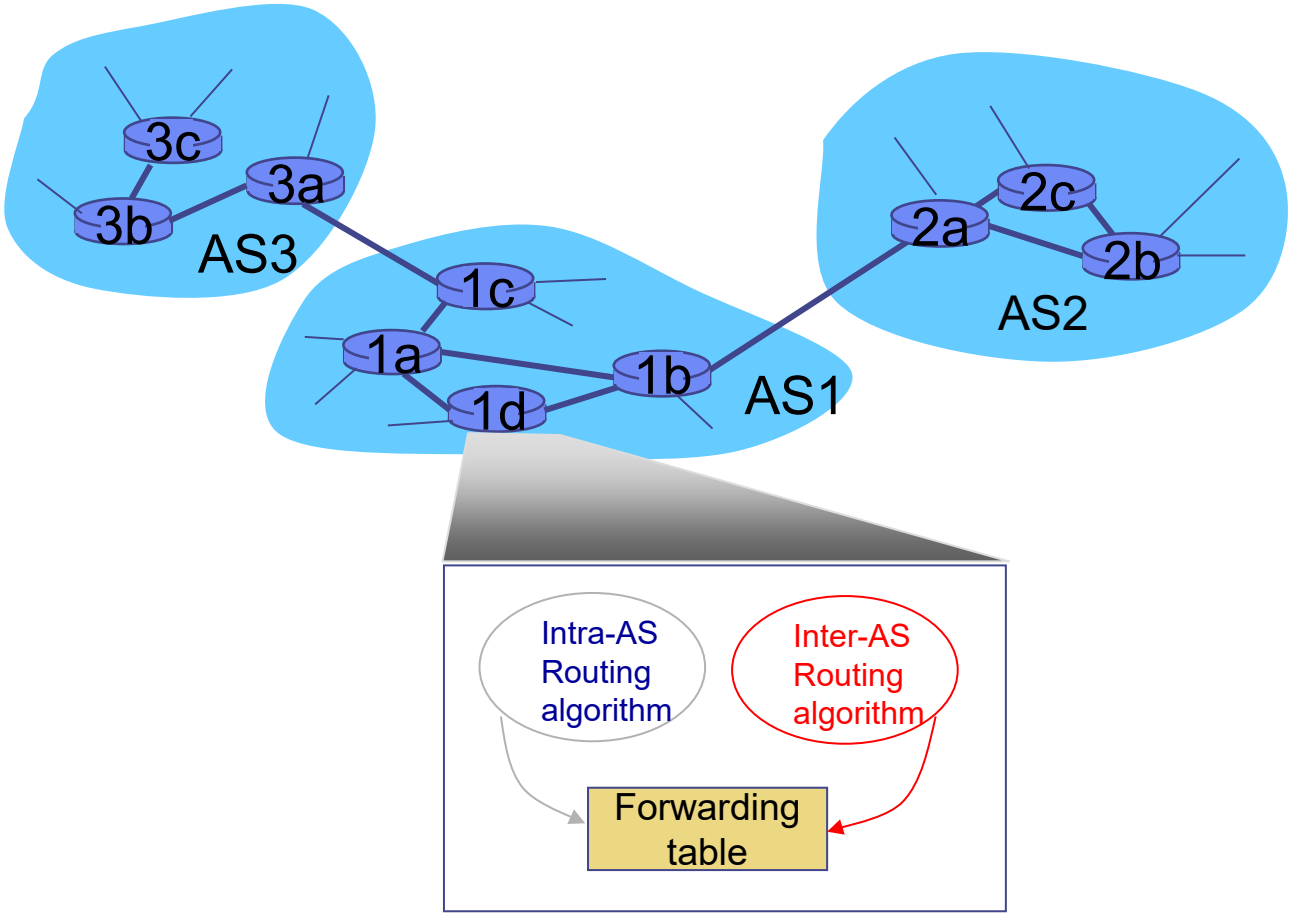
Inter-AS routing

- Routing among AS'es
- Gateways perform inter-domain routing (as well as intra-domain routing)

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Interconnected AS's

- Forwarding table configured by both intra- and inter-AS routing algorithm
  - intra-AS routing determine entries for destinations within AS
  - inter-AS & intra-AS determine entries for external destinations

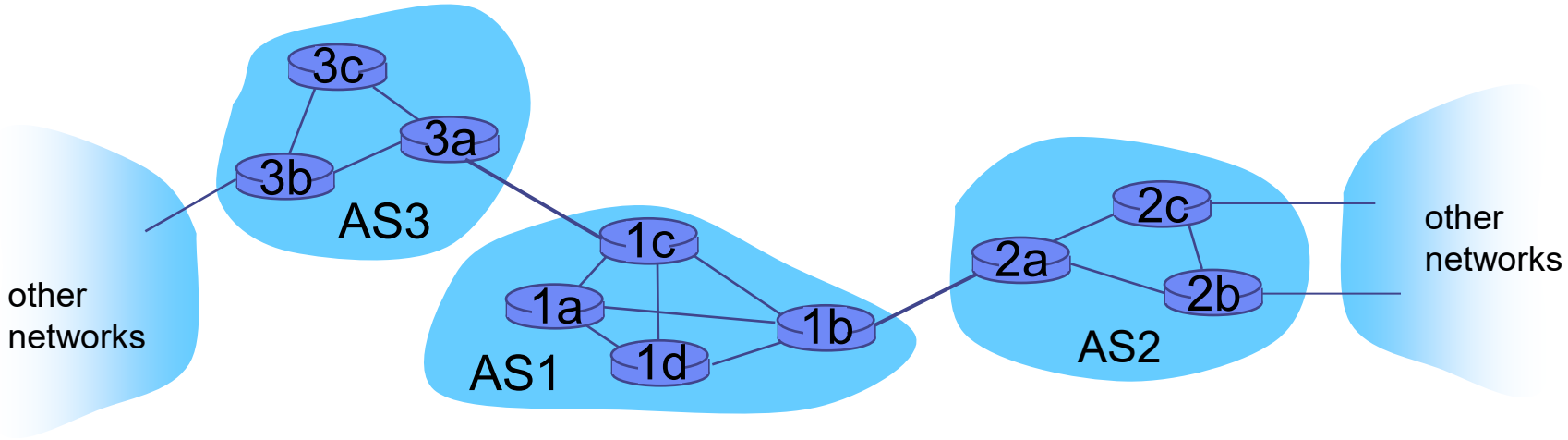


Inter-AS tasks

- Router in AS1 receives datagram destined outside of AS1:
  - Router should forward packet to gateway router, but which one?

AS1 must:

1. Learn which destinations are reachable through AS2, which through AS3
2. Propagate this reachability info to all routers in AS1



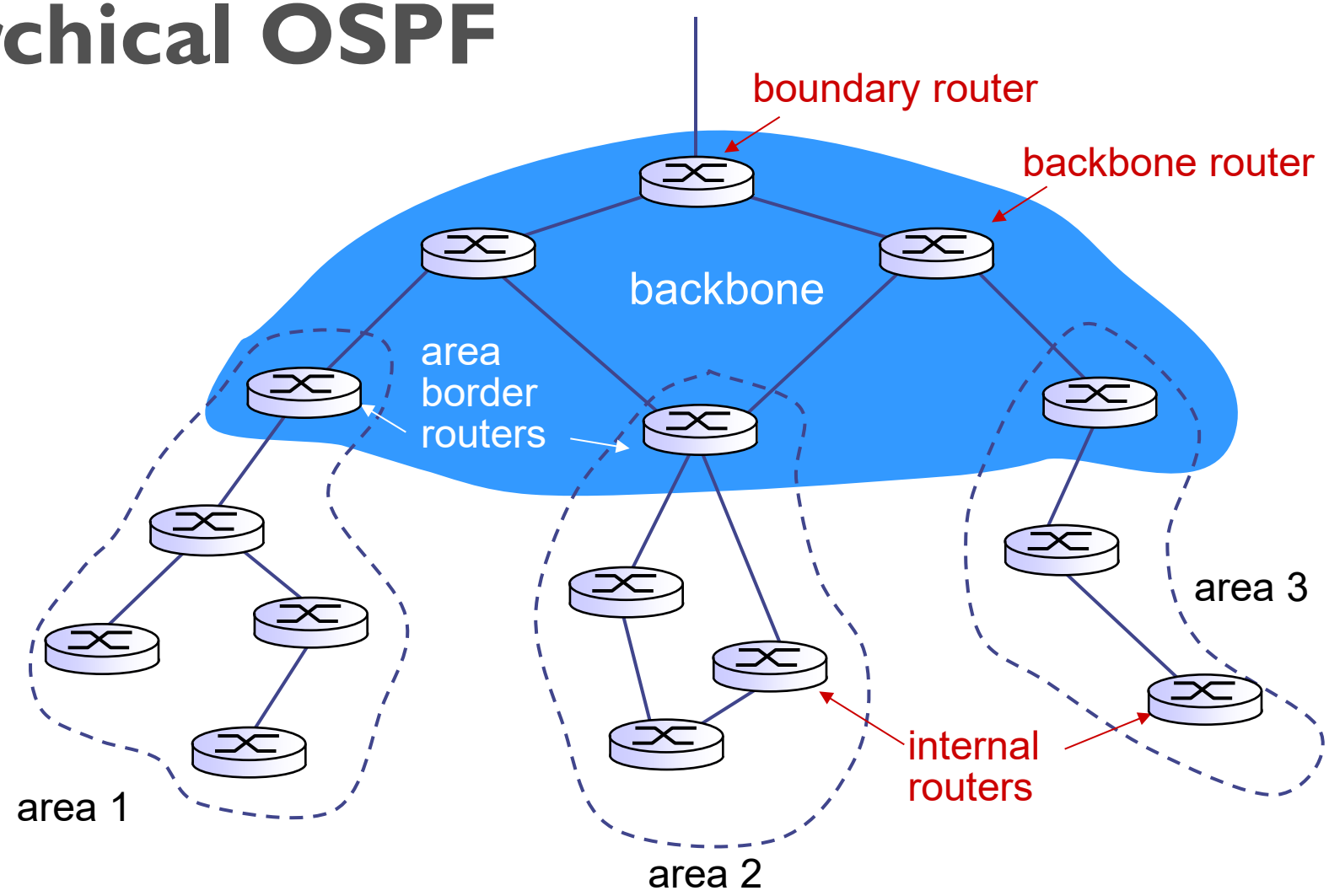
Intra-AS Routing

- Also known as *interior gateway protocols (IGP)*
- Most common intra-AS routing protocols:
  - RIP: Routing Information Protocol [RFC 1723]
    - classic DV: DVs exchanged every 30 secs
    - no longer widely used
  - EIGRP: Enhanced Interior Gateway Routing Protocol
    - DV based
    - formerly Cisco-proprietary for decades (became open in 2013 [RFC 7868])
  - OSPF: Open Shortest Path First [RFC 2328]
    - link-state routing
    - IS-IS protocol (ISO standard, not RFC standard) essentially same as OSPF

OSPF (Open Shortest Path First)

- Widely used
- Open: publicly available
- Uses link-state algorithm
  - link state packet dissemination
  - topology map at each node
  - route computation using Dijkstra's algorithm
- Router floods OSPF link-state advertisements to all other routers in *entire* AS
  - carried in OSPF messages directly over IP (rather than TCP or UDP)
  - link state: for each attached link

Hierarchical OSPF



8.3 Inter Autonomous System Routing

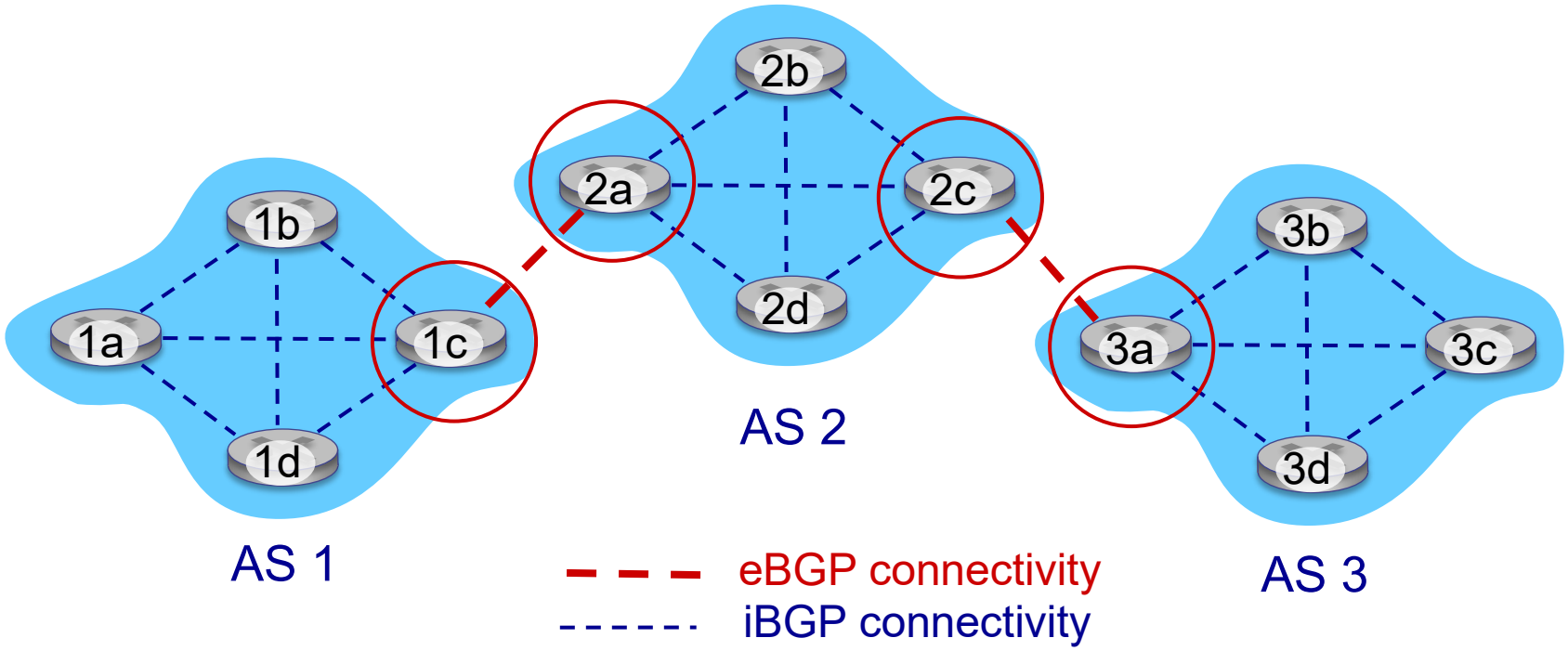
Internet inter-AS routing: BGP

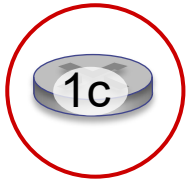
BGP (Border Gateway Protocol):
The de facto inter-domain routing protocol "glue that holds the Internet together"

- BGP provides each AS a means to:
  - Obtain subnet reachability information from neighboring As'es **(eBGP)**
  - Determine routes to other networks based on reachability information and **policy**
  - Propagate reachability information to all AS-internal routers **(iBGP)**
- Allows subnet to advertise its existence to rest of Internet: **"I am here"**

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e(xternal)BGP, i(nternal)BGP connections



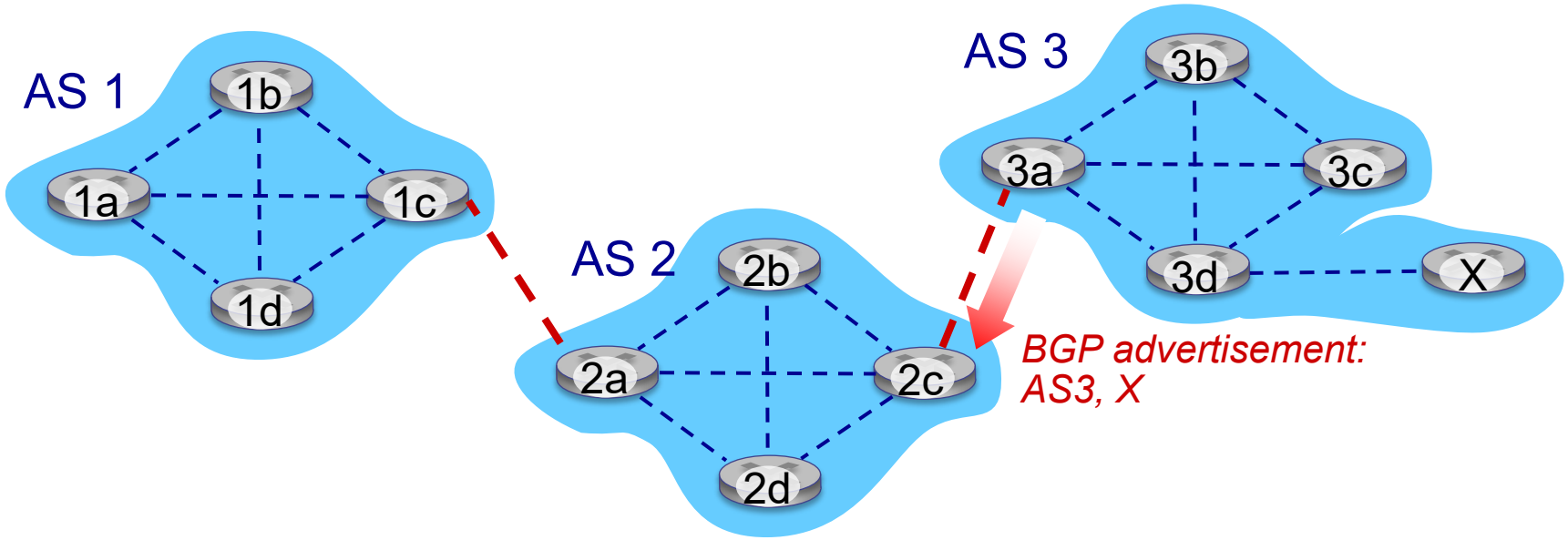


gateway routers run both eBGP and iBGP protocols

BGP basics

- **BGP session:** two BGP routers ("peers") exchange BGP messages over TCP:
  - advertising *paths* to different destination network prefixes (BGP is a "path vector" protocol)

- When AS3 gateway router 3a advertises path AS3,X to AS2 gateway router 2c:
  - AS3 **promises** to AS2 it will forward datagrams towards X



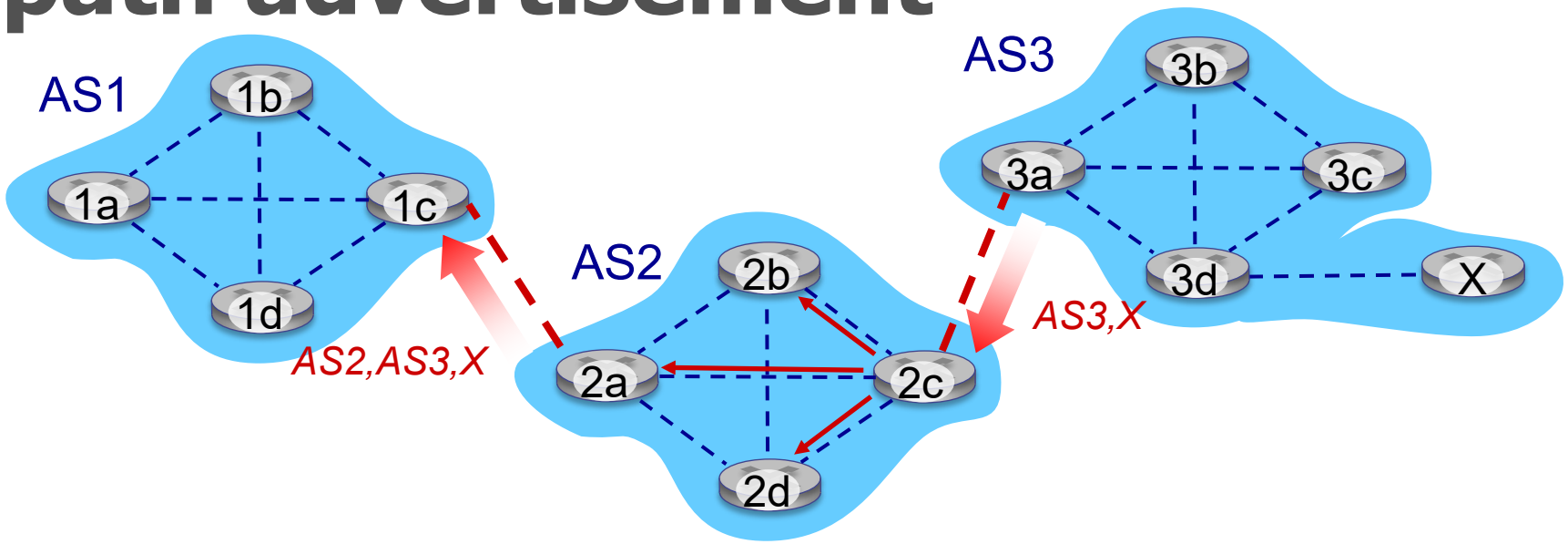
Path Attributes and BGP routes

- BGP advertised attributes: prefix + attributes = "route"
  - Prefix: destination
  - Two important attributes:
    - **AS-PATH:** list of AS's through which prefix advertisement has passed
    - **NEXT-HOP:** indicates specific internal-AS router to next-hop AS

- **Policy-based routing:**
  - Gateway receiving route advertisement uses **import policy** to accept/decline path (e.g., never route through AS Y).
  - AS policy also determines whether to **advertise** path to other other neighboring AS'es

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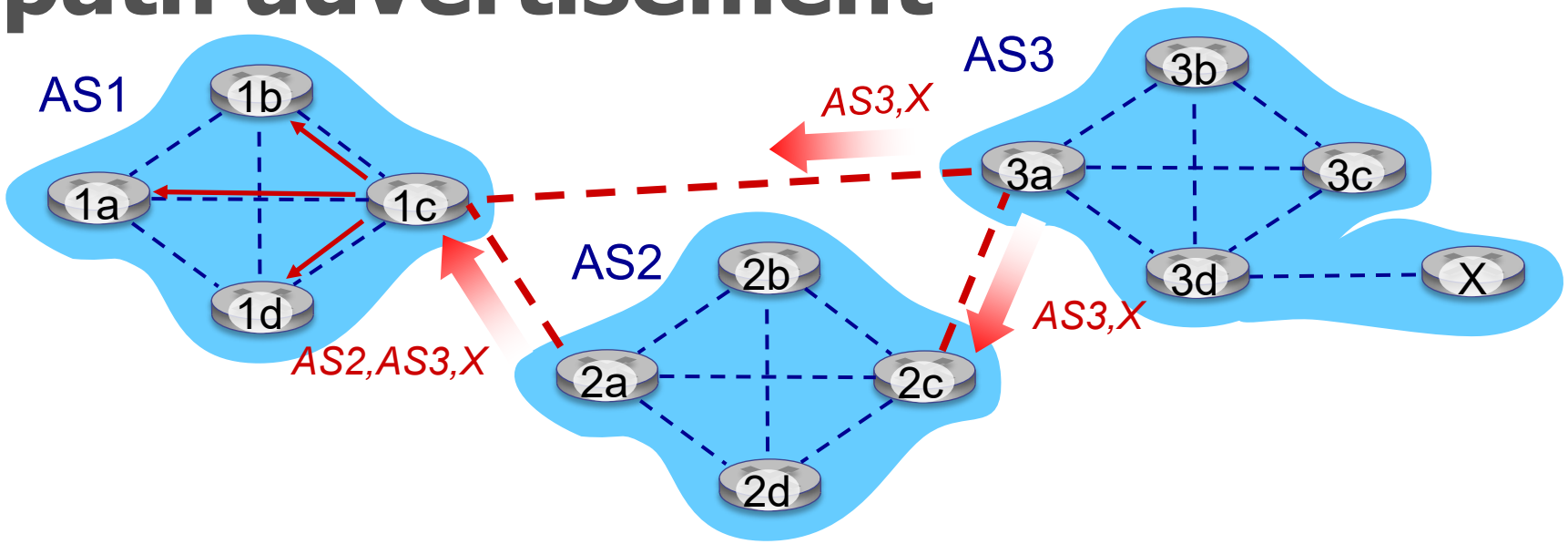
BGP path advertisement



- AS2 router 2c receives path advertisement AS3,X (via eBGP) from AS3 router 3a
- Based on AS2 policy, AS2 router 2c accepts path AS3,X, propagates (via iBGP) to all AS2 routers
- Based on AS2 policy, AS2 router 2a advertises (via eBGP) path AS2, AS3, X to AS1 router 1c

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BGP path advertisement

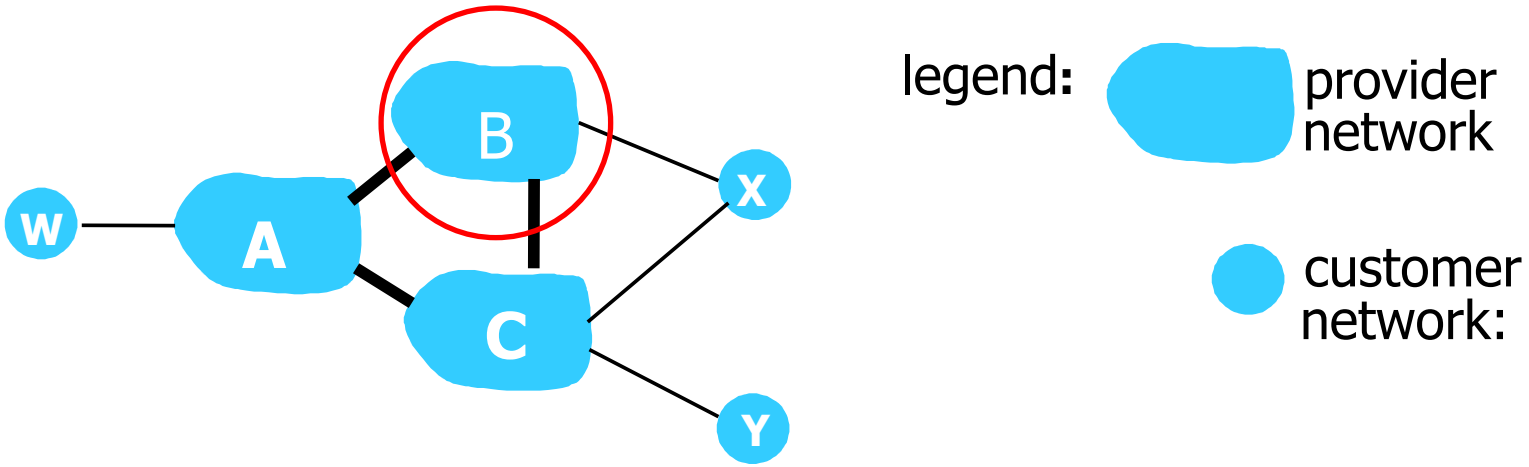


gateway router may learn about multiple paths to destination:

- AS1 gateway router 1c learns path *AS2,AS3,X* from 2a
- AS1 gateway router 1c learns path *AS3,X* from 3a
- Based on policy, AS1 gateway router 1c chooses path *AS3,X, and advertises path within AS1 via iBGP*

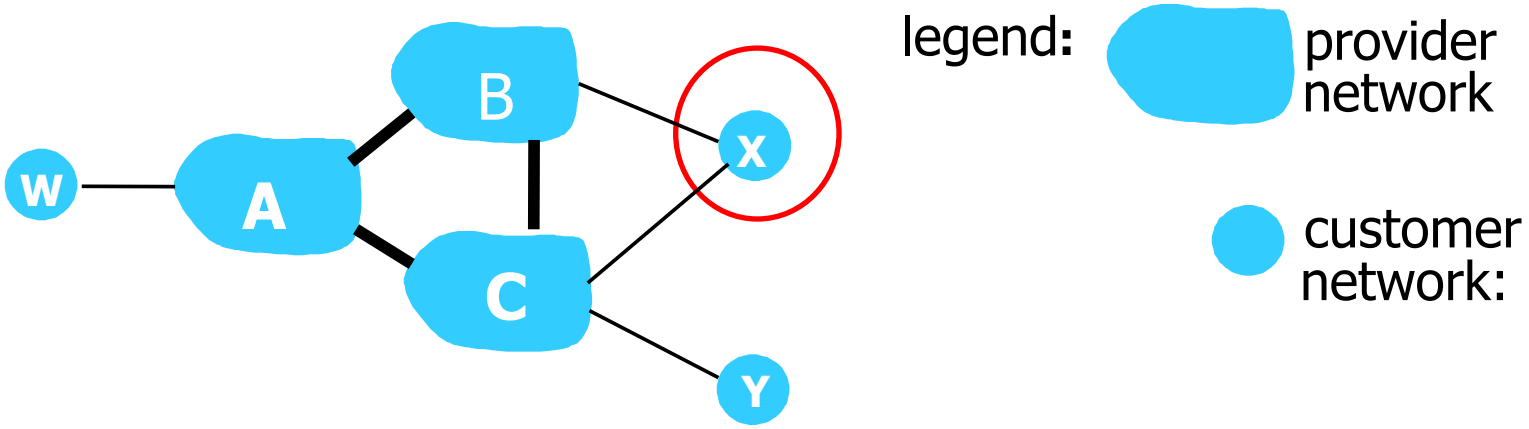
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BGP: achieving policy via advertisements



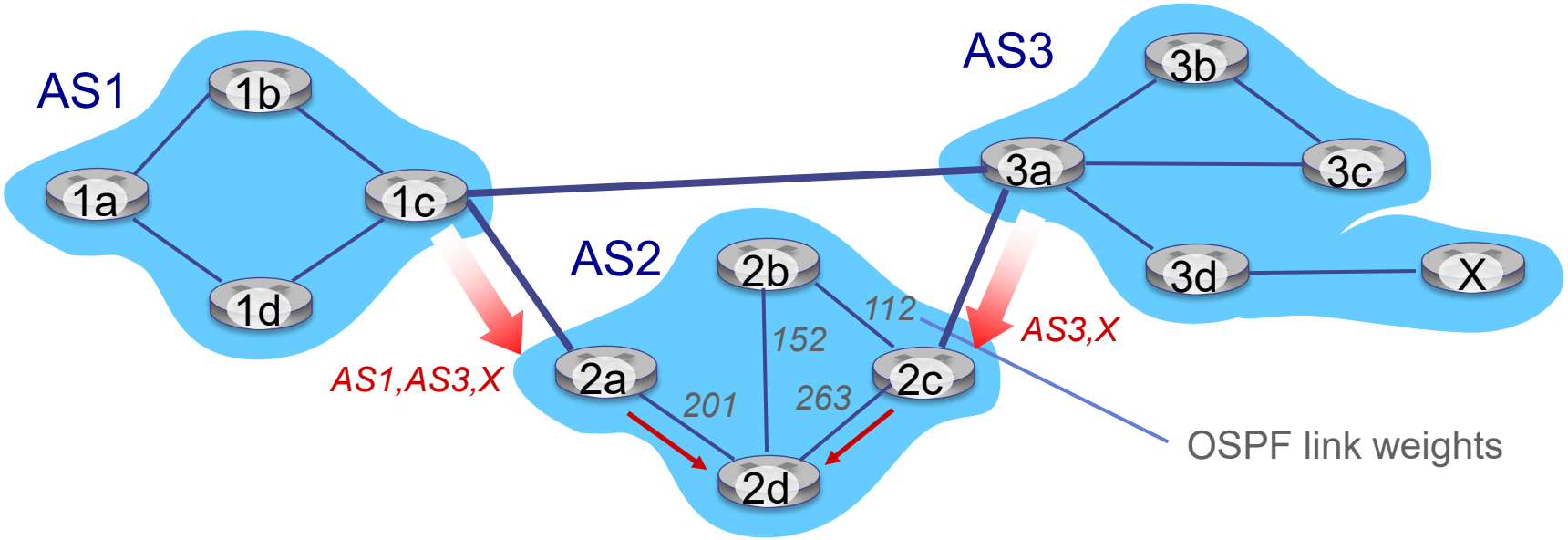
An ISP (B) may only wants to route traffic to/from its customer networks (does not want to carry transit traffic between other ISPs)

BGP: achieving policy via advertisements



Customer network x is dual-homed
A customer network may not want to route traffic between ISP

Hot Potato Routing



- 2d learns (via iBGP) it can route to X via 2a or 2c
- Choose local gateway that has least intra-domain cost
  - (e.g., 2d chooses 2a, even though more AS hops to *X*): don't worry about inter-domain cost!

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Why different Intra-, Inter-AS routing ?

Policy:

- Inter-AS: administrate control over its traffic is routed (who routes through its net)
- Intra-AS: single admin, so no policy decisions needed

Scale:

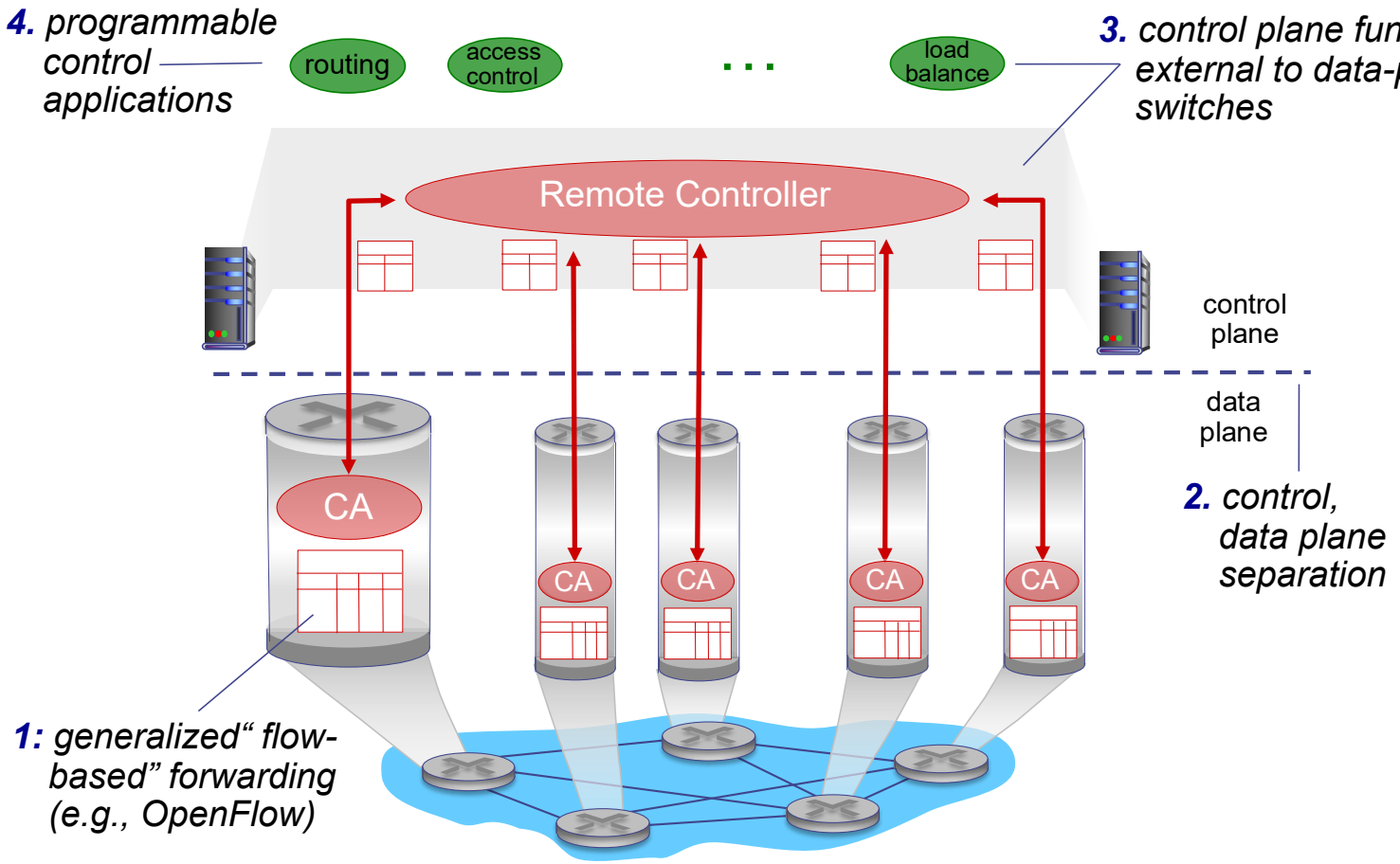
- Hierarchical routing saves table size, reduced update traffic
  - Route to subnets instead of all adresses

Performance:

- Intra-AS: can focus on performance
- Inter-AS: policy may dominate over performance

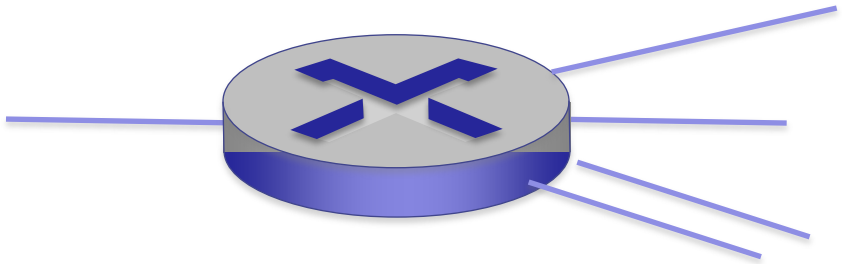
8.4 Software Defined Networking & Generalized Forwarding in Brief

Software defined networking (SDN)



OpenFlow data plane abstraction

- A protocol for SDN controllers to program their network switches
- *Flow*: defined by header fields
- Generalized forwarding: simple packet-handling rules
  - *Pattern:* **match** values in packet **header fields**
  - *Actions: for matched packet:* drop, forward, modify, matched packet or send matched packet to controller
  - *Priority*: disambiguate overlapping patterns
  - *Counters:* #bytes and #packets

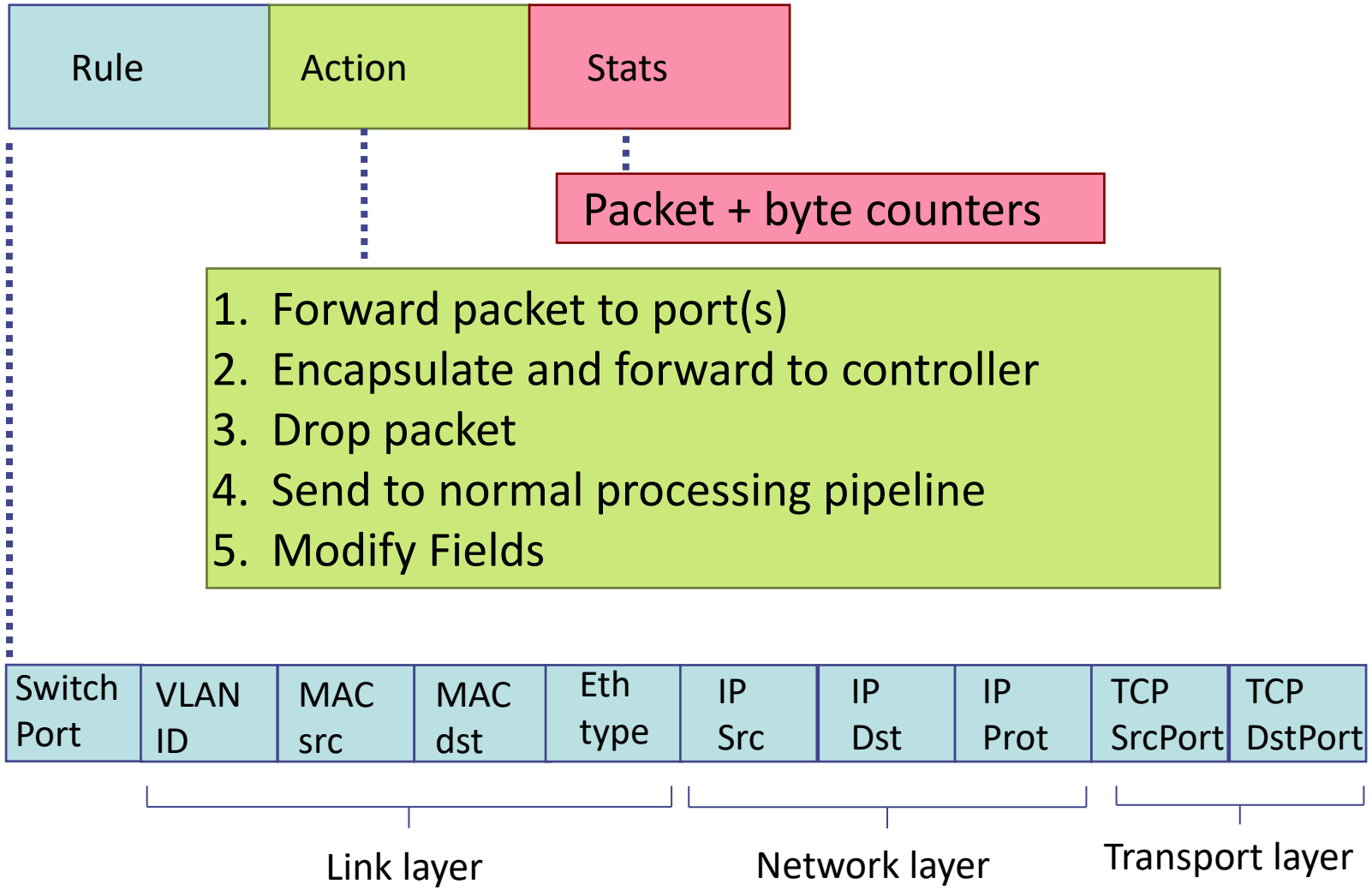


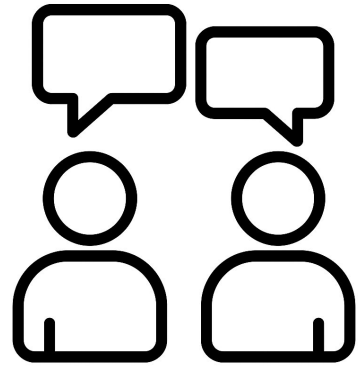
* : wildcard

1. src=1.2.*.*, dest=3.4.5.* → drop
2. src = *.*.*.*, dest=3.4.*.* → forward(2)
3. src=10.1.2.3, dest=*.*.*.* → send to controller

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OpenFlow: Flow Table Entries





Discuss pros and cons

8.5 Internet Control Message Protocol

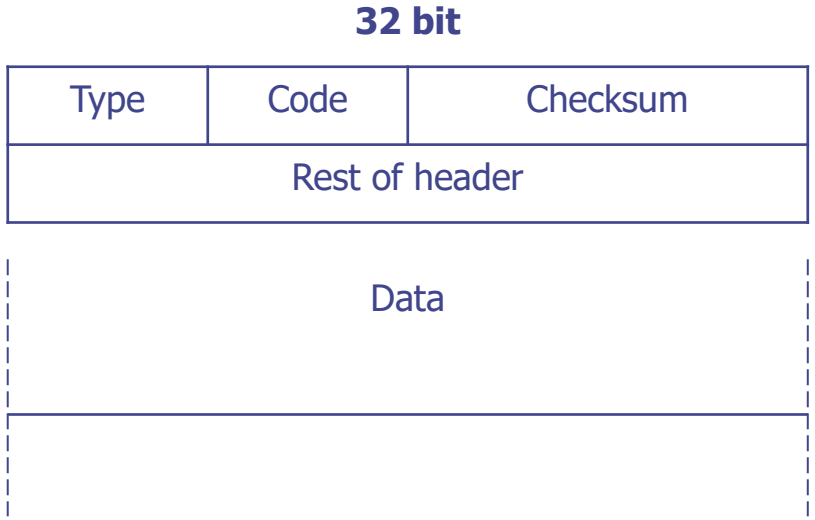
ICMP: Internet Control Message Protocol

- Used by hosts & routers to communicate network-level information
  - Error reporting: unreachable host, network, port, protocol
  - Echo request/reply (used by ping)
- ICMP msgs carried in IP datagrams
  - Protocol number 1
- ICMP message:
  - type, code, checksum
  - For error messages it also includes some of the problem causing IP datagram (header + 8 bytes of payload)

Type	Code	description
0	0	echo reply (ping)
3	0	dest. network unreachable
3	1	dest. host unreachable
3	2	dest. protocol unreachable
3	3	dest. port unreachable
3	6	dest. network unknown
3	7	dest. host unknown
4	0	source quench
8	0	echo request (ping)
9	0	route advertisement
10	0	router discovery
11	0	TTL expired
12	0	bad IP header

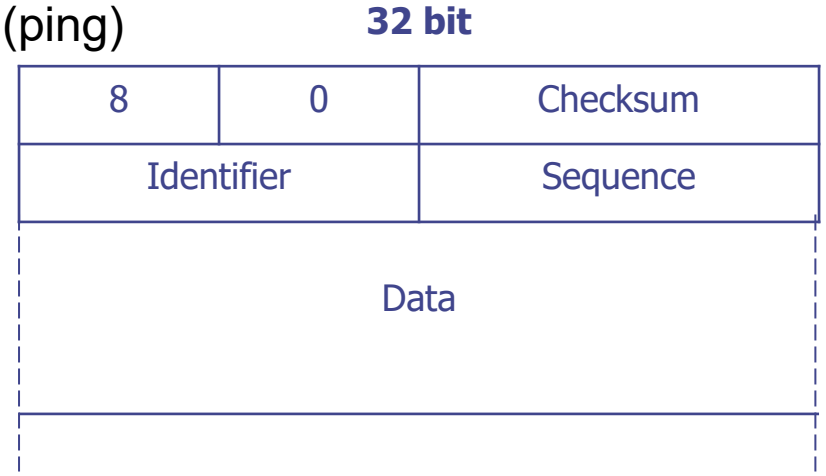
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ICMP: Internet Control Message Protocol

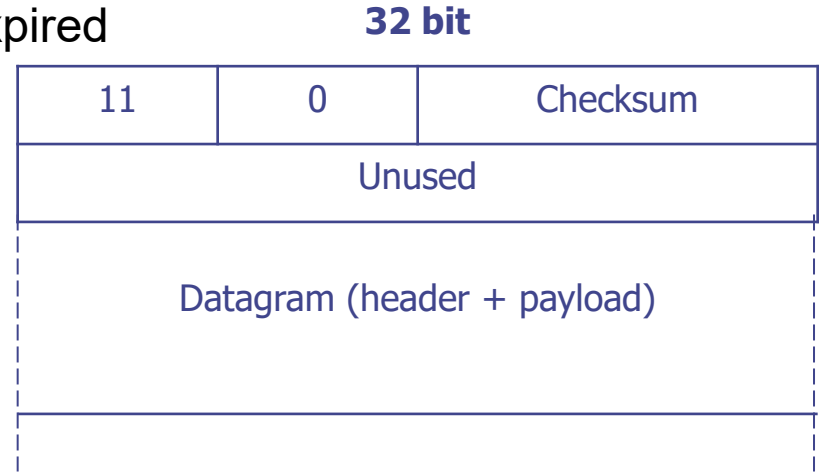


Examples:

Eg. Type = 8.
echo request (ping)



Eg. Type = 11
TTL expired



Traceroute and ICMP

- Source sends series of ICMP packets (or UDP segments) to destination
  - first set has TTL =1
  - second set has TTL=2, etc.
  - unlikely port number
- When datagram in n^{th} set arrives to n^{th} router:
  - router discards datagram and sends source ICMP message (type 11, code 0)
  - ICMP message include name of router & IP address

- When ICMP message arrives, source records RTTs

Stopping criteria:

- packet eventually arrives at destination host
- Destination returns ICMP "port unreachable" message (type 3, code 3)
- Source stops

